



Opacity and Explainability

5ARF0

Guest Lecture

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Plan

What is **opacity**, why does it occur in contemporary ML-powered AI, and why is it a problem?

How can **explainability** be increased, whom does it help, and why?

How does explainability help **align AI systems with human norms**?

ML-powered AI

An AI system is trained to solve a **difficult problem**.

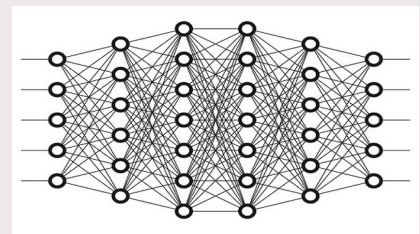
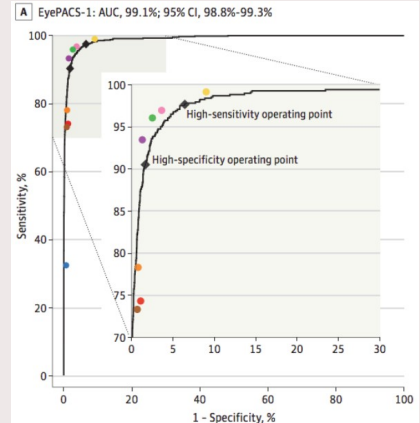
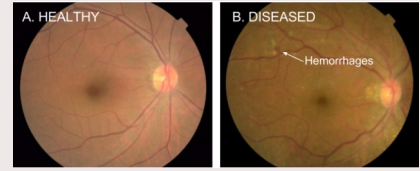
→ E.g. diagnosing diabetic retinopathy

Training involves applying a **learning algorithm** to a (large) dataset.

→ E.g. gradient descent over retinal image database, ca. 120K images

The learning algorithm fine-tunes parameters of a (complex) **neural network** until it performs well.

→ E.g. 22M parameter convolutional neural network



ML-powered AI

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Problem domain is ill-understood.

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Humans don't tell the system how to solve the problem. The system learns on its own.

The learning algorithm fine-tunes parameters of a (complex) **neural network** until it performs well.

→ E.g. 22M parameter convolutional neural network

The system is high-dimensional and nonlinear.

ML-powered AI

Problem domain is ill-understood.

The system may find solutions that are surprising, hard to interpret, or beyond established theory.

Humans don't tell the system how to solve the problem. The system learns on its own.

Solutions can be particularly intricate.

The system is high-dimensional and nonlinear.

ML-powered AI

The system may find solutions that are surprising, hard to interpret, or beyond established theory.

Solutions can be particularly intricate.

We can now solve problems that we could not solve previously, even in ill-understood domains.

Problem domain is ill-understood.

Humans don't tell the system how to solve the problem. The system learns on its own.

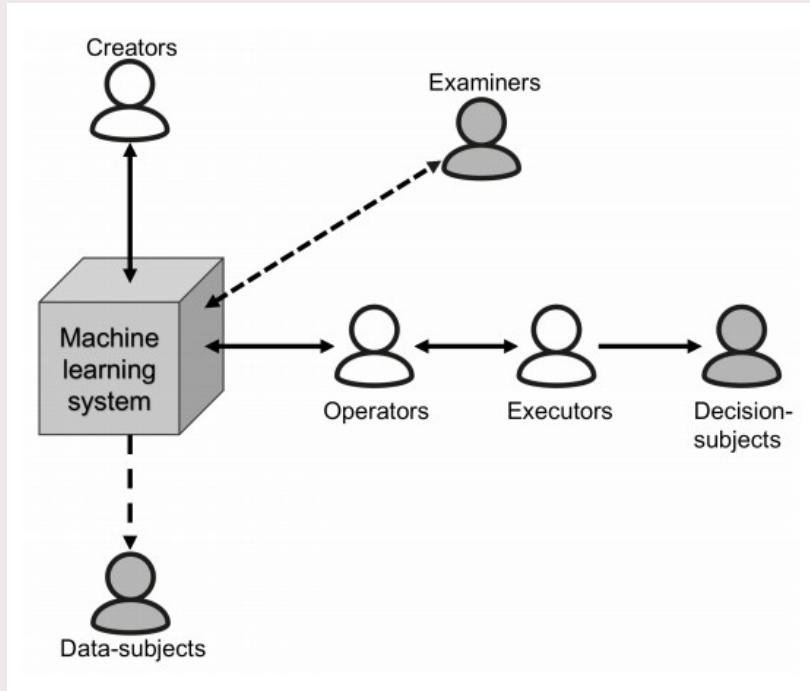
The system is high-dimensional and nonlinear.

Opacity

Unfortunately, the advantages of ML-powered AI come with the disadvantage that this kind of AI is characteristically **opaque**.

Opacity: Stakeholders do not possess the information they need to **effectively interact** with an AI system.

Opacity



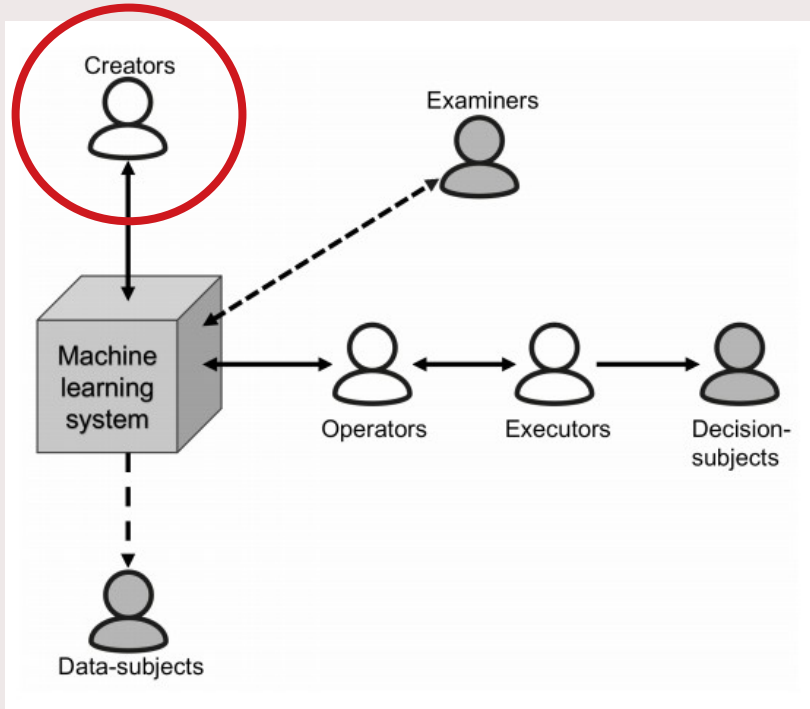
Many different **stakeholders** inhabit the “ML ecosystem”.

Each stakeholder **interacts** with an AI system in characteristic ways.

In order to interact effectively, they need relevant **understanding** of the system.

Opacity prevents this kind of understanding.

Opacity

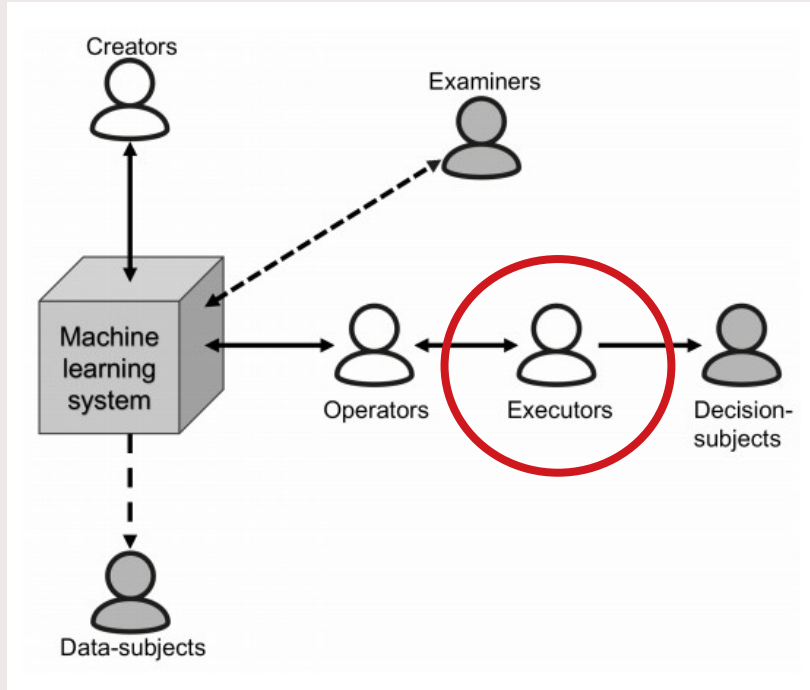


E.g. **software developers** need to design, train, debug...

To do so, they need to **understand the causal dependencies** between system architecture, parameters, etc. and system behavior: "How does the system work?"

The system's **internal complexity and learning** (rather than hand-coding) prevents this.

Opacity

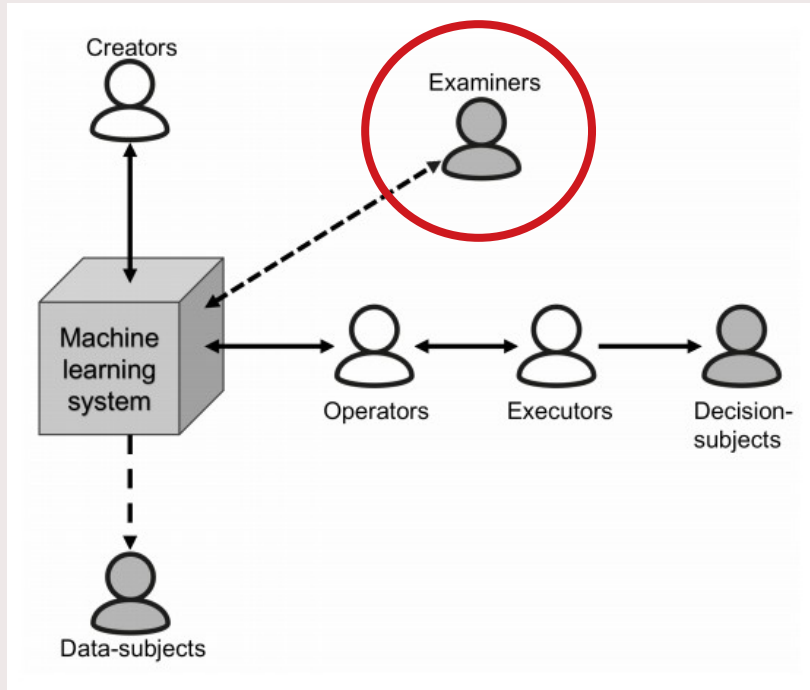


E.g. **bank employees** rely on AI systems to make decisions or predictions and identify patterns in data.

To do so, they must be able to **interpret** the system's behavior in context: "Why does the system do what it does?"

This is difficult if the system's behavior is **hard to understand**, either because it is too surprising or intricate, or because we do not understand the domain.

Opacity



E.g. **regulators** need to evaluate system safety and conformity with laws, predict system failures and attribute responsibility...

To do so, they must be able to compare the system's behavior and functioning **against relevant norms**: “Does the system do what it should be doing?”

This is difficult if the learned solution is **intricate** and **hard to interpret**.

Opacity

Opacity is **ethically relevant** because it can lead to poor performance and lack of oversight.

Opacity is **not just** an ethical issue; it is also inefficient and impractical.

Opacity is **stakeholder-relative**: Whether a system is opaque or not, and what the consequences of opacity are, depends on who you are and in what way you interact with AI.

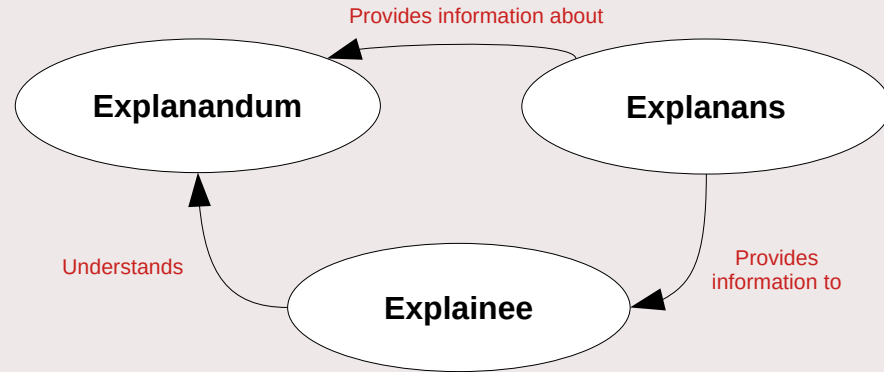
Presumably, **solutions** to the problems that opacity causes will likewise be stakeholder-relative.

Explainability

If lack of understanding is the problem, provision of understanding through **explanation** might be considered a solution.

Explanation:

the act of providing information (the *explanans*) about something (the *explanandum*) to someone (the *examinee*) so as to increase their **understanding**.



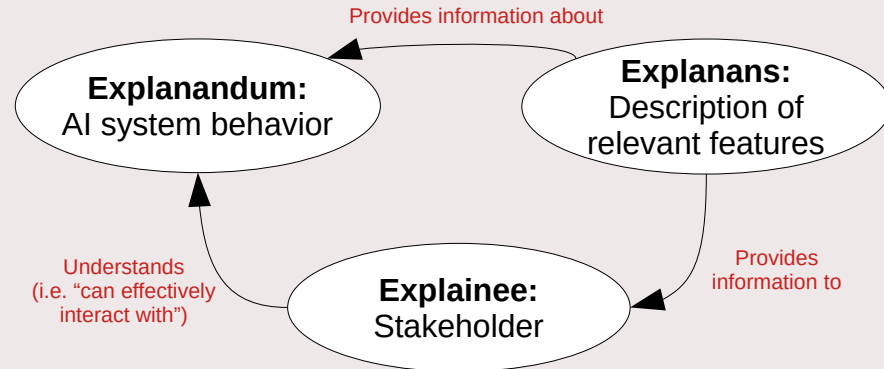
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An AI system becomes **explainable** to a stakeholder if they can understand it in a relevant way.



Explainability

Within the field of **Explainable AI (XAI)**, we can observe three basic approaches to promoting explainability:

- Inherently explainable systems

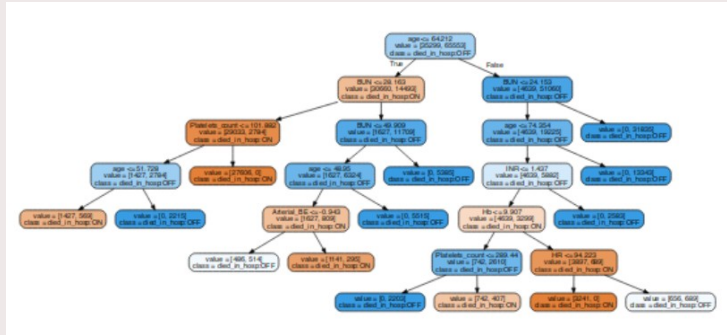
- Self-explaining systems

- Post-hoc explanation methods

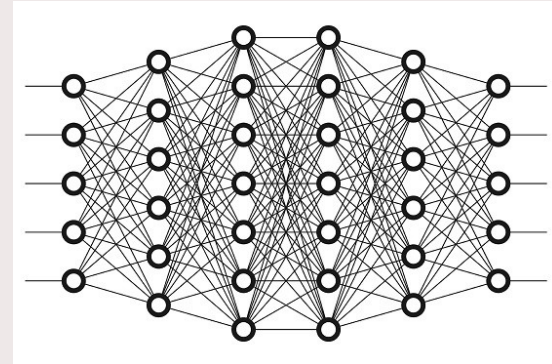
Inherently Explainable Systems

Inherently explainable systems (sometimes: “interpretable” or “glass-box” systems) avoid opacity by using ML methods that promote solutions are less intricate and less difficult to understand.

E.g. decision trees vs. neural networks



VS.



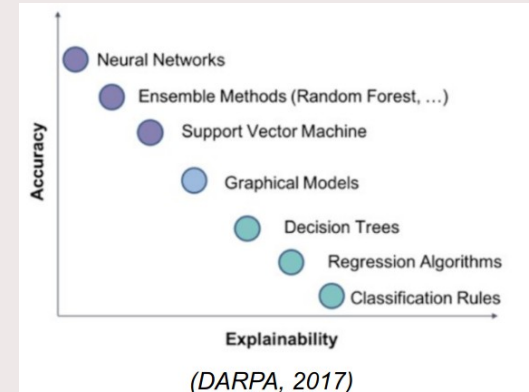
Inherently Explainable Systems

But, some concerns:

Is there an **accuracy/explainability tradeoff**?

Do e.g. decision trees perform less well (for some kinds of problems) than e.g. neural networks?

How explainable are they really? Does it help to use decision trees, if individual nodes are themselves difficult to map onto meaningful concepts?

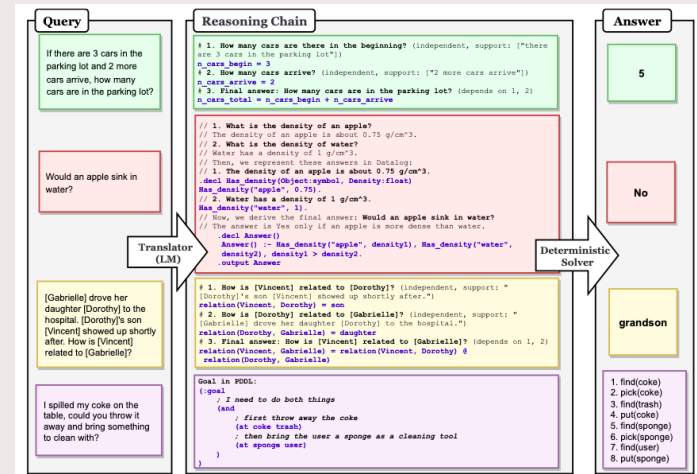


Self-Explaining Systems

Self-explaining systems aim to overcome opacity by generating explanatory information alongside the actual system outputs.

E.g. Asking an LLM to provide **“reasons”** for its outputs.

E.g. **“chain of thought”** reasoning in LLMs, generating output of comprehensible reasoning chains that culminate in the final prediction, answer, or result.



Self-Explaining Systems

The current challenge is to ensure that the reported reasoning is **veridical rather than fabricated**, i.e. that it is an accurate representation of the process that actually produced the system's outputs, rather than just another output that provides a plausible, but non-actual explanation.

Self-generated explanations are no more reliable than **human self-reports**.

Post Hoc Explanation

Post hoc explanations combat opacity through statistical, mathematical, or computational methods that describe relevant features of opaque AI systems and their behaviors.

There is a **great variety of these methods**, they work in different ways, describe different kinds of features, and thus, are useful for different purposes and to different stakeholders.

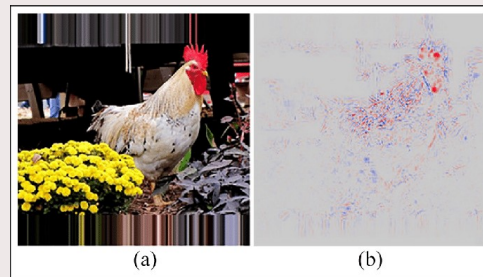
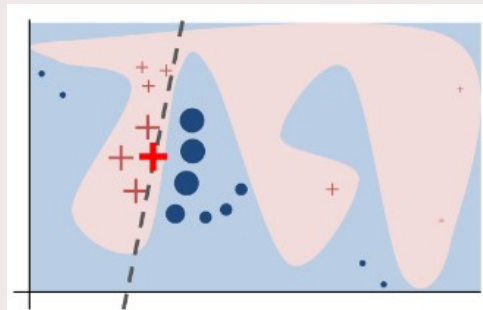
Post Hoc Explanation

Linear approximation methods (such as *LIME*) can (locally) approximate an opaque system's behavior, allowing us to predict its decision just on the basis of a few inputs.

E.g. “If you are between 25 and 30 years old, getting married will result in a positive loan decision”

Saliency maps (e.g. generated by *LRP*) help us know which features of a system's input most determine the system's output.

“E.g. the system recognizes the chicken mostly by the features on its head”



Post Hoc Explanation

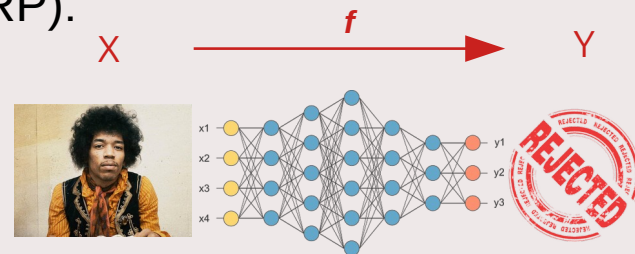
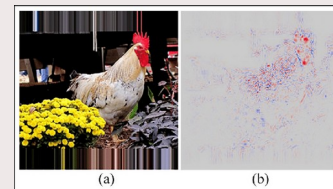
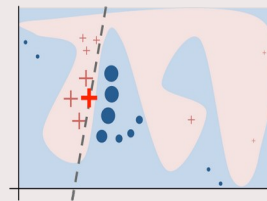
These methods **describe AI systems' behaviors** locally and/or globally.

“What the AI system is doing”

E.g. *Linear approximators* (e.g. LIME):
what is function f (for a limited domain)?

E.g. *Feature-importance measures* (e.g. SHAP, LRP):
what was X , for a given Y ?

E.g. *Counterfactual explanations*:
what would X have to be, to yield Y ?



Post Hoc Explanation

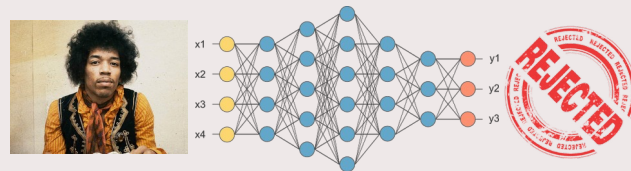
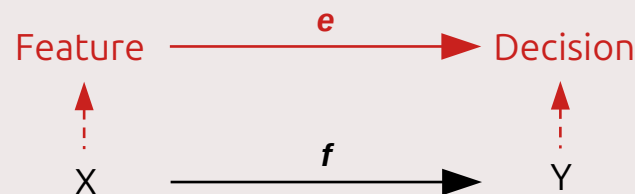
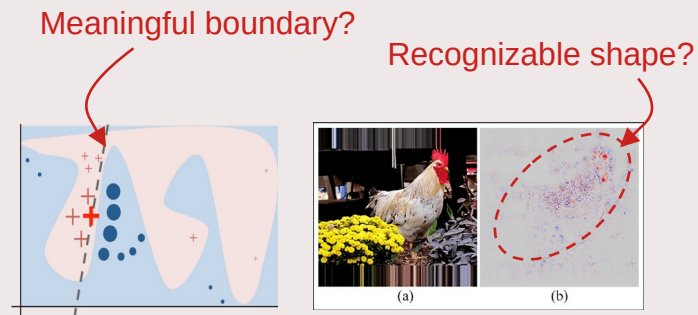
If we are lucky, these descriptions can be **interpreted** using familiar concepts.

Does the system's behavior track a regularity or feature in the environment?

“Why the AI system does what it does”

Of course, we are **not always so lucky**.

The system might be tracking a very intricate and hard-to-describe feature.



Post Hoc Explanation

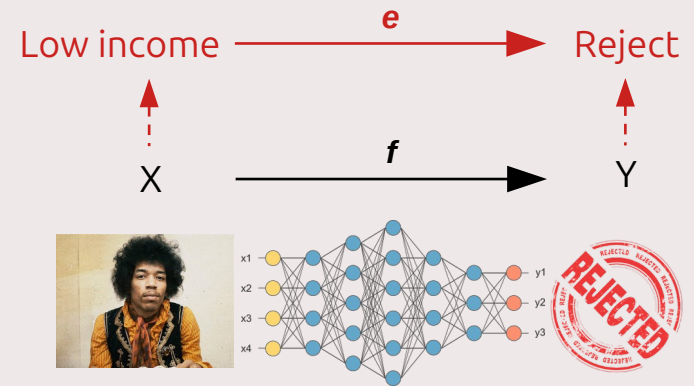
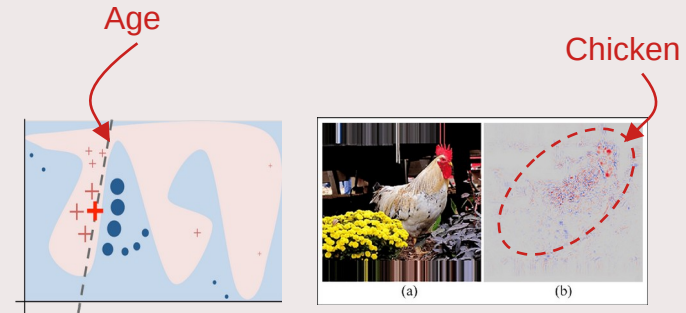
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Does the system's behavior track a regularity or feature in the environment?

“Why the AI system does what it does”

Thus interpreted, these methods can be used to **understand** the system's solution in context.

“The system makes loan decisions on the basis of **income level**.”



Post Hoc Explanation

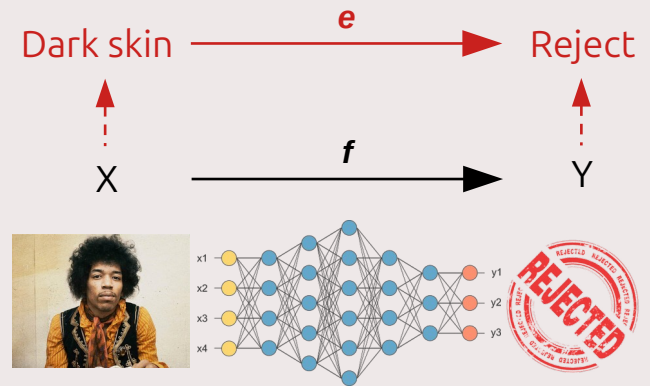
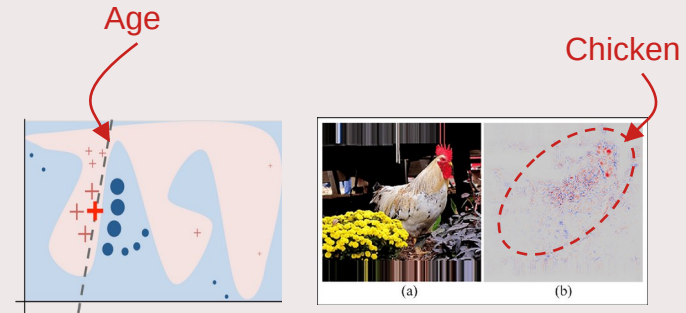
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Thus interpreted, these methods can be used to **understand** the system's solution in context.

“The system makes loan decisions on the basis of **skin color**.”



Post Hoc Explanation

Explanations that tell us “what a system is doing” may be useful for **end-users** who need to know what kinds of inputs to provide to a system to get desired outputs.

Interpretations that tell us “why a system is doing what it does” are most useful for **domain-experts** (such as decision-makers) who need to put the AI system’s behavior in context.

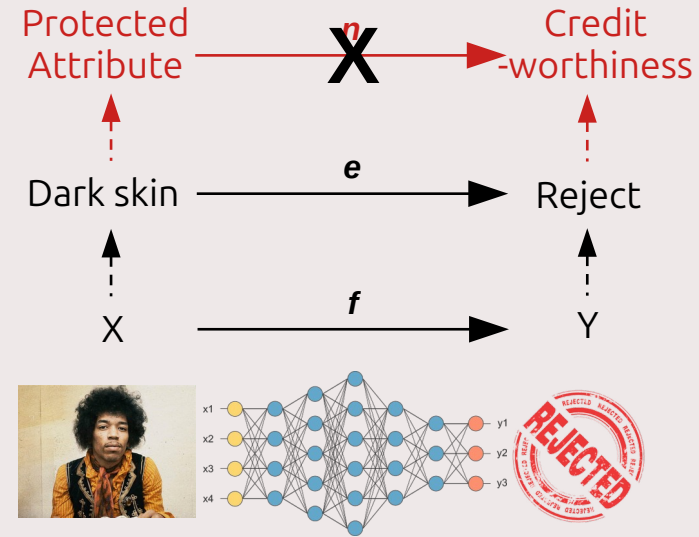
Unfortunately, even if we can often explain, **we may still not be able to interpret**, because AI systems will have learned to track environmental regularities that are too intricate, obscure or unfamiliar to be described using available concepts.

Post Hoc Explanation for Alignment

Interpreted descriptions may also allow us to **compare AI systems' behaviors against relevant norms**.

E.g. financial, legal, and ethical norms of credit-scoring.

The concepts used to interpret (descriptions of) a system's behavior must “fit” the concepts used in the specification of relevant norms.

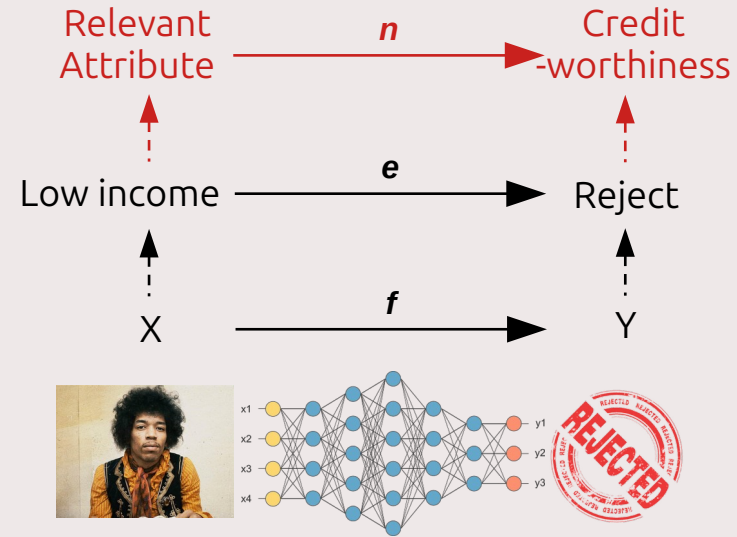


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Post Hoc Explanation for Alignment

AI systems whose behaviors track relevant norms are **well-aligned**.

Alignment $\approx$ AI systems **do what we want them to do**.

Post-hoc explanations help us **assess** an AI system's alignment insofar as their behavior can be explained, interpreted, and compared to a relevant norm.

In these sense, these explanations can be useful for **regulators**.

Limitations of Post Hoc Explanation

Unfortunately, these post hoc methods have some characteristic **limitations**.

Lack of interpretability: Not all features and regularities can be interpreted.

Lack of robustness: Different explanation methods may suggest different explanations.

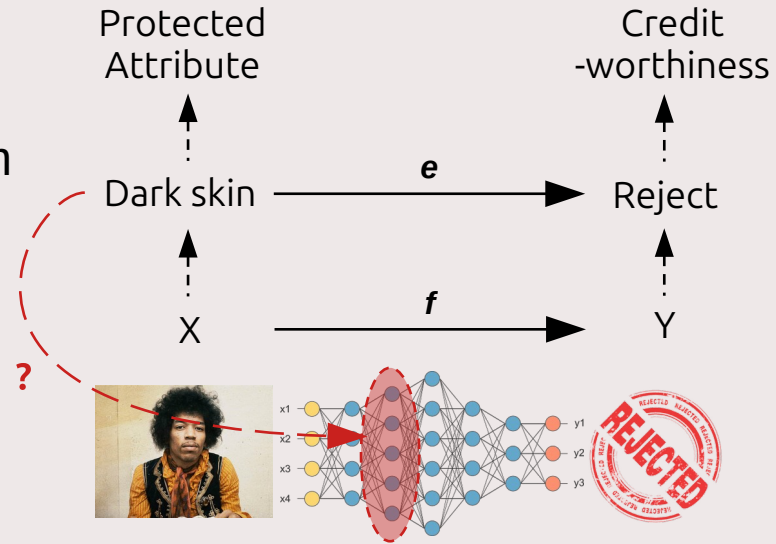
Limitations of Post Hoc Explanation

A deeper limitation of these methods is that they provide limited information about the underlying **causal structures** of the system.

Although we may know what an AI system is doing and why, we still do not know how it works.

As a result, they provide limited support for **intervention**.

This is often needed by **developers**.

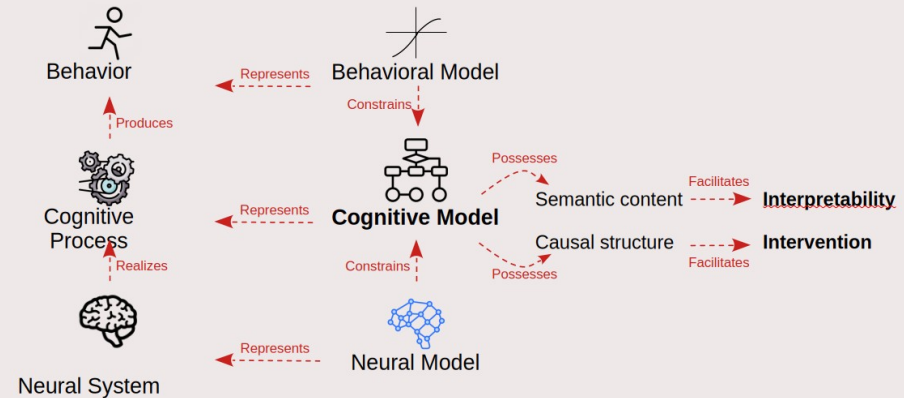


Cognitivist Explanation

Recent work in explainable AI aims to describe and predict not just AI systems' behavior, but also the **causal mechanisms** and **representational structures** that underlie that behavior.

This work mirrors the explanatory logic of human psychology:
a “**cognitivist explanation**”

Cognitivist explanations center on the development of **cognitive models**.



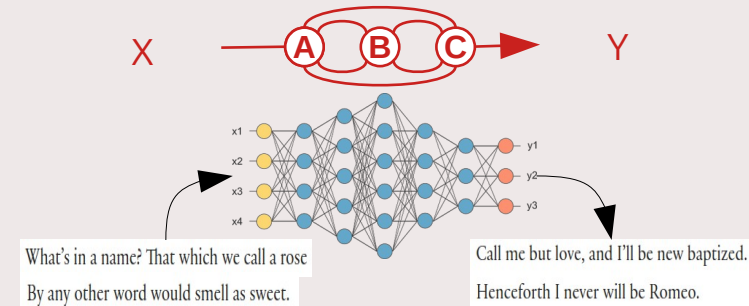
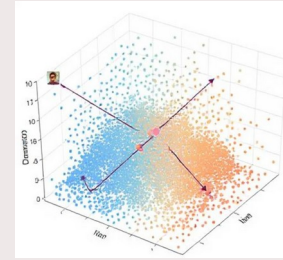
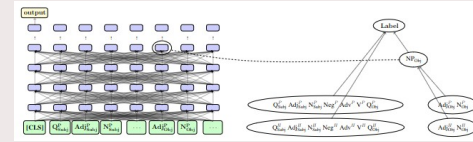
Cognitivist Explanation

A growing number of methods **describe mechanisms** that mediate between input and output.

“How the system works”

E.g. *causal abstraction analysis* (CAA)
a causal model that reproduces
transformation of X to Y

E.g. *latent space analysis*
a projection of activation values
while X is being transformed to Y



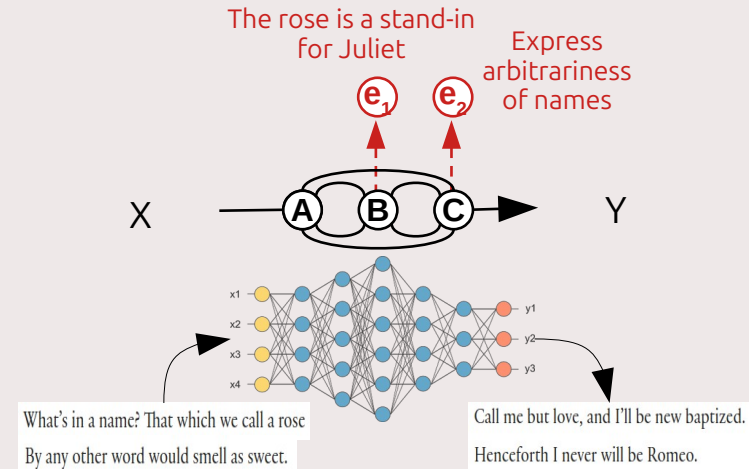
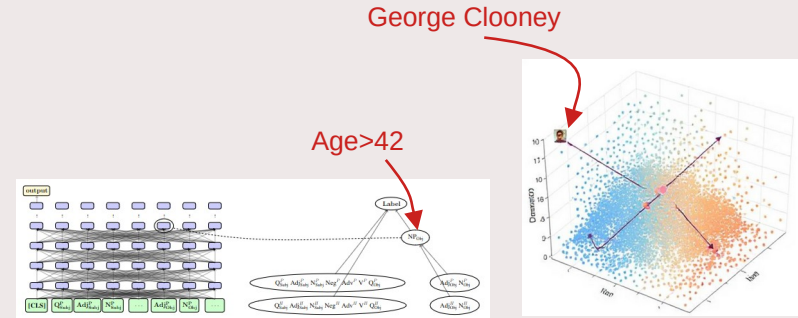
Cognitivist Explanation

Sometimes, the variables of these mechanisms can be **interpreted** using relevant concepts: Does the system **represent** a recognizable regularity or feature?

“Why the system does what it does”

Causal nodes can be interpreted?

Embedding vectors of meaningful concepts?



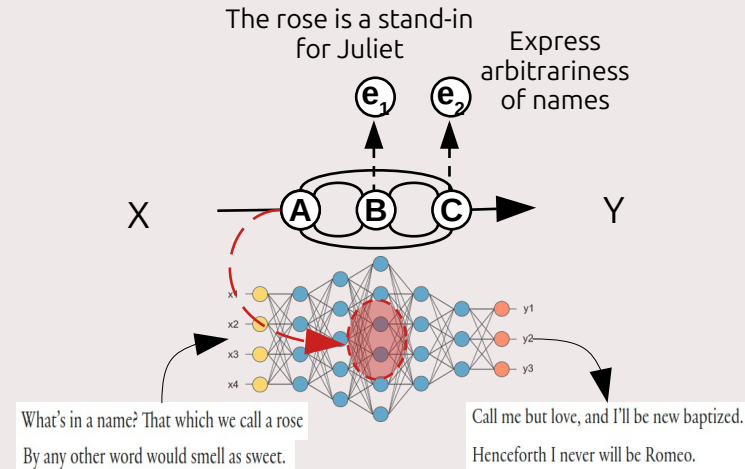
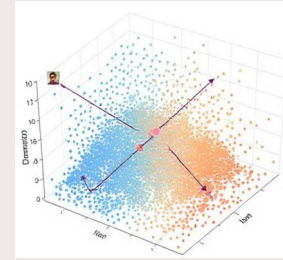
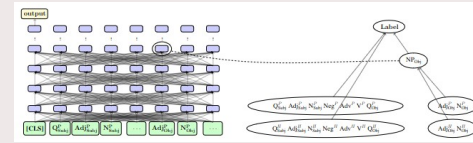
Cognitivist Explanation

Ideally, these variables can also be identified with structural features of the system to support targeted **interventions**.

“Can we make the system work better?”

Alignment of causal nodes with system components.

Localization of embedding vectors.

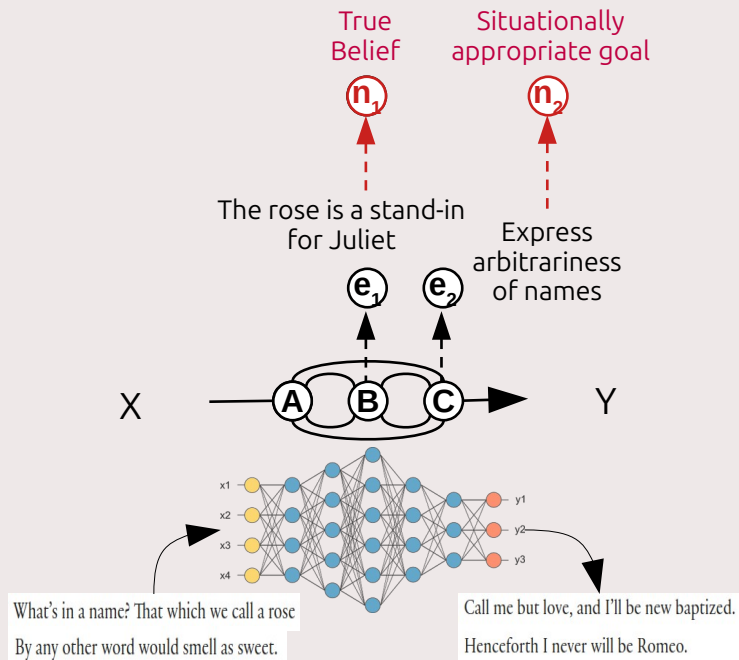


Cognitivist Explanation for Deep Alignment

Mechanism-based explanations help us evaluate a system's alignment by **comparing internal representations** against relevant norms.

Does the system have true beliefs?

Is the system pursuing the right goals?

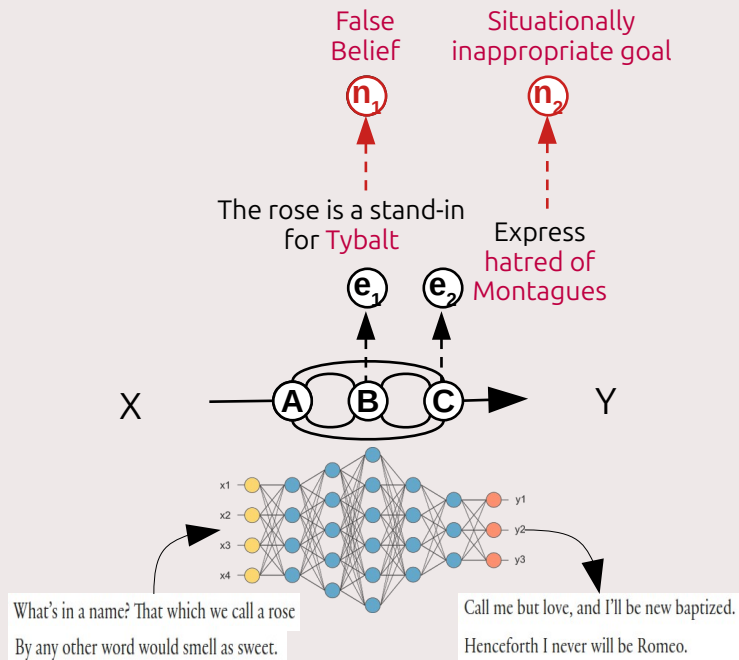


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Cognitivist Explanation for Deep Alignment

AI systems whose internal representations track normatively relevant features are **deeply aligned**.

Deep Alignment $\approx$ AI systems “**think**” what we want them to think.

Cognitivist explanations help us **assess** an AI system’s deep alignment insofar as their “thoughts” can be explained, interpreted, and compared to a relevant norm.

They also help us **ensure** deep alignment by specifying targets for intervention on the AI system’s internal mechanisms, thereby satisfying the requirements of **developers**.

Cognitivist Explanation: Examples and Limitations

Cognitivist explanation is in many ways the “**cutting edge**” of explainable AI.

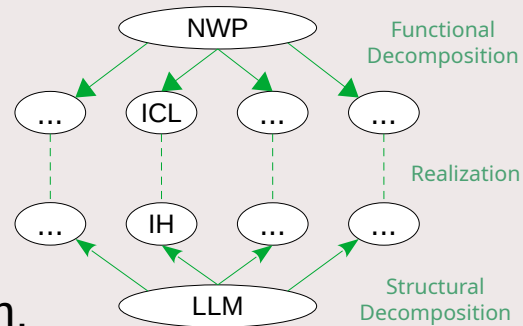
Tools and methods exist, but these are still **limited**.

Cognitivist Explanation: Examples and Limitations

E.g. Mechanistic Interpretability

Decompose behavior (e.g. next-word prediction) into functional parts (e.g. in-context-learning).

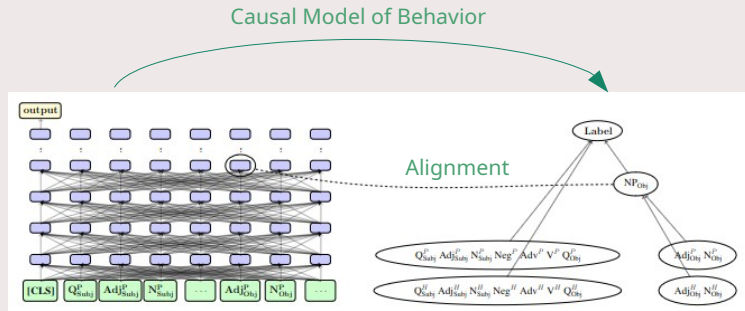
Identify mechanisms for each functional part (e.g. induction heads), check for presence in system.



Mirrors a **functional analysis** explanatory strategy from psychology.

But, remains unclear whether the functional parts identified using this method can be **interpreted**.

Cognitivist Explanation: Examples and Limitations



E.g. Causal Abstraction Analysis

Construct a causal model that reproduces the system's behavior.

Align model variables with system elements (e.g. learned representations).

Resultant model is simple and reproduces global behavior, but also supports **systematic interventions** on that behavior.

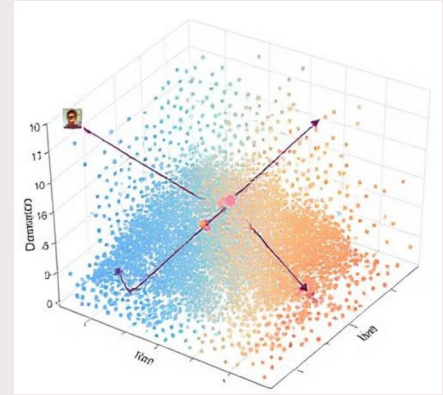
But, it remains unclear whether the method **scales** to more complex systems.

Cognitivist Explanation: Examples and Limitations

E.g. Latent space analysis

Describe a projection of an AI system's activation values while processes certain inputs.

Associate vectors within the space with particular concepts.



The projection helps **interpret** a system's “thoughts”, but it is not always clear to what extent it actually plays a **causal role** in the generation of outputs.

Summary

ML-driven AI is powerful, but characteristically **opaque**.

Opacity is a **stakeholder's lack of understanding** of the AI system, which leads to their inability to effectively interact with it.

Practical harms to many different stakeholders!

Explanations may be able to provide this kind of understanding.

Three different approaches: inherent, self-explaining, post hoc.

Summary

Inherently explainable systems may be limited by the accuracy/explainability tradeoff.

Self-explaining systems may fabricate and only give the appearance of explanation.

Asking an LLM “why it does what it does” is not particularly helpful.

Summary

Traditional **post hoc explanations** can tell us “what” and AI system is doing and “why”.

Useful for end-users, domain-experts e.g. decision-makers.

Not always possible to interpret, and no support for interventions.

They can also help us compare AI systems’ behaviors to relevant norms, and therefore help us assess these systems’ behavioral **alignment**.

Summary

More cutting edge **cognitivist explanations** can tell us “how” AI systems work, by describing interpreted representational structures that (ideally) figure in causal processes.

Useful for developers.

Difficult to scale, no guarantee that interpretable structures can actually be found.

They can also help us compare AI systems’ “thoughts” to relevant norms, and therefore help us assess these systems’ **deep alignment**.